

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I _____vanumu surya satvika(192524254)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: v.surya satvika

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:

Question 1: Water Jug Problem (4-Gallon and 3-Gallon Jugs)

Problem :

Two jugs are available:

- Jug A = 4 gallons
- Jug B = 3 gallons

Operations allowed:

- Fill a jug completely.
- Empty a jug completely.
- Pour water from one jug to another until one becomes empty or the other becomes full.

Goal:

Obtain exactly **2 gallons** in the **4-gallon jug**.

State Space Representation:

Represent each state as:

(A, B)

where

- A = amount of water in 4-gallon jug
- B = amount of water in 3-gallon jug

Initial State:

(0,0)

Goal State:

(2,x)

where x can be any value.

Solution Steps:

Step	Action	State (4G,3G)
Initial	Both empty	(0,0)
1	Fill 3-gallon jug	(0,3)
2	Pour into 4-gallon jug	(3,0)
3	Fill 3-gallon jug again	(3,3)
4	Pour into 4-gallon until full	(4,2)
5	Empty 4-gallon jug	(0,2)
6	Pour remaining 2 gallons into 4-gallon jug	(2,0)

State representation:

(0,0)

↓

(0,3)

↓

(3,0)

↓

(3,3)

↓

(4,2)

↓

(0,2)

↓

(2,0) ← Goal State

Search Strategy Used:

This problem can be solved using **Breadth-First Search (BFS)** because BFS explores all possible states level by level and guarantees the shortest sequence of actions to reach the goal state.

Algorithm:

- Start
- Initialize (0,0)
- Repeat
- Fill a jug
- OR
- Empty a jug
- OR
- Transfer water between jugs
- Generate new state
- If state = (2,0)
- Stop
- Else
- Continue
- End

Interpretation of Result:

The objective is successfully achieved. After six valid operations, the **4-gallon jug contains exactly 2 gallons of water**, satisfying the required goal state. This demonstrates how a state-space search approach can solve real-world AI problem-solving tasks using legal operators.

Question 2: Mars Rover – Intelligent Agent Analysis

The Mars Rover is an intelligent autonomous agent that explores the surface of Mars, collects scientific data, analyzes rocks and soil, and sends the information back to Earth. It must operate independently because communication with Earth has a significant time delay.

(i) What are the percept's for this agent?

Percepts are the inputs received by the Mars Rover through its sensors.

- Images captured by cameras
- Rock and soil composition data
- Temperature readings
- Atmospheric pressure
- Radiation levels
- Terrain obstacles (rocks, slopes, craters)
- GPS/Navigation position
- Battery level
- Wheel status

(ii) Characteristics

Environment Property	Description
Observable	Partially Observable (cannot see the entire environment at once)
Agents	Single Agent
Deterministic	Mostly Stochastic (uncertain terrain and weather)
Episodic/Sequential	Sequential
Static/Dynamic	Dynamic
Discrete/Continuous	Continuous
Known/Unknown	Partially Known

(iii) Actions the Agent Can Take

The Mars Rover can perform the following actions:

- Move forward
- Move backward
- Turn left
- Turn right
- Stop
- Capture images
- Drill rocks
- Collect soil samples
- Analyze collected samples
- Transmit data to Earth
- Recharge using solar panels
- Avoid obstacles
- Plan a new route

(iv) Performance Evaluation

The Mars Rover's performance is measured by:

- Number of successful samples collected
- Accuracy of scientific observations
- Distance explored
- Safe obstacle avoidance
- Battery efficiency
- Successful communication with Earth
- Completion of assigned mission
- Reliability of collected data

(v) Suitable Agent Architecture and Justification:

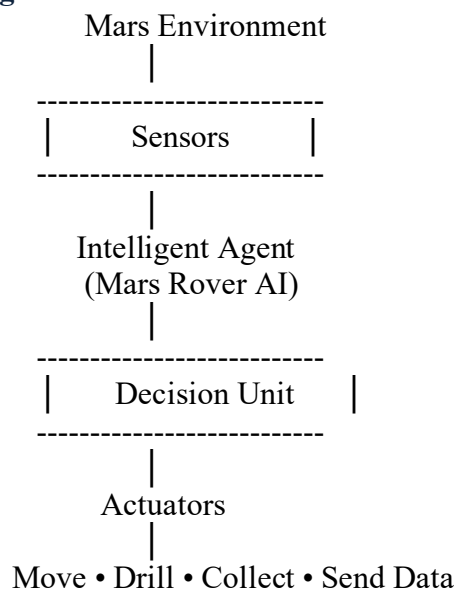
Justification:

- Learns from previous experiences.
- Selects the safest and most efficient path.
- Maximizes scientific discoveries while minimizing energy consumption.
- Adapts to unknown terrain and unexpected obstacles.
- Makes autonomous decisions when communication with Earth is delayed.

PEAS Representation

Component	Description
Performance Measure	Accurate sample collection, safety, battery efficiency, successful exploration
Environment	Mars surface, rocks, dust, craters, hills, atmosphere
Actuators	Wheels, robotic arm, drill, cameras, antenna
Sensors	Cameras, temperature sensor, GPS, spectrometer, radiation sensor, obstacle detector

Simple Diagram:



Interpretation of Result:

The Mars Rover is an autonomous intelligent agent that continuously senses its environment, processes information, makes decisions, and performs actions without direct human control.

Question 3: 8-Queens Problem – Search Space and Problem Formulation

Problem Statement:

Place **8 queens** on an **8 × 8 chessboard** such that no two queens attack each other.

This means that no two queens can be in the same **row, column, or diagonal**.

Problem Understanding:

The objective is to find a valid arrangement of eight queens where every queen is safe from attack by the other seven queens. This is a classical Artificial Intelligence search problem that can be solved using state-space search and constraint satisfaction techniques.

Problem Formulation:

1. Initial State

An empty chessboard with no queens placed.

```
□□□□□□□□
□□□□□□□□
□□□□□□□□
□□□□□□□□
□□□□□□□□
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□□□□□□□□
□□□□□□□□
```

2. State Space

A state represents the current arrangement of queens on the chessboard.

Example State:

```
Q □ □ □ □ □ □
□ □ □ □ Q □ □ □
□ □ □ □ □ Q □
□ Q □ □ □ □ □
□ □ □ □ □ Q □
□ □ Q □ □ □ □
□ □ □ □ □ □ Q
□ □ □ Q □ □ □
```

Each new state is generated by placing one additional queen.

3. Actions (Operators)

The possible actions are:

- Place a queen in an empty row.
- Ensure the queen is **not** in the same:
 - Row

- Column
 - Diagonal
- Continue until all 8 queens are safely placed.

4. Goal Test

The goal is achieved when:

- All **8 queens** are placed.
- No two queens attack each other.

5. Path Cost

Each queen placement has an equal cost.

Total Cost = **8 placements**

Search Strategy

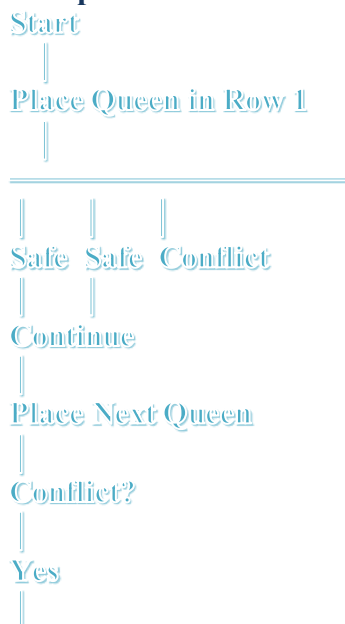
The 8-Queens problem can be solved using:

- Depth First Search (DFS)
- Backtracking Search (**Most Suitable**)
Backtracking places queens one by one. If a conflict occurs, it removes the last queen and tries another position.

Backtracking Algorithm

1. Start
2. Place Queen in Row 1
3. For each next row
4. Check safe position
5. If safe
6. Place Queen
7. Else
8. Backtrack
9. Repeat until 8 queens are placed
10. Stop

Search Space Illustration:





Advantages of Backtracking

- Reduces unnecessary searching.
- Eliminates invalid configurations quickly.
- Efficient for constraint satisfaction problems.
- Guarantees finding a valid solution if one exists.

Interpretation of Result

The 8-Queens problem demonstrates how AI uses **state-space search** and **constraint satisfaction** to solve complex problems. By using **Backtracking Search**, invalid arrangements are discarded early, allowing the algorithm to efficiently find a valid configuration where all eight queens are safely placed.

Question 4: OLA Cab Booking – Problem Solving Agent and Pseudocode

Problem Statement:

A customer wants to travel from one location to another using the **OLA Cab Booking** mobile application. The customer can choose different cab types such as **Mini, Micro, Sedan, Shared, Prime**, etc., based on comfort and budget. Identify the suitable problem-solving agent and write the pseudocode.

Problem Understanding

The OLA Cab Booking application acts as an **Intelligent Problem-Solving Agent**. It receives the customer's request, searches for available cabs, compares different options, selects the best cab based on the user's preference, and books the ride.

Suitable Problem-Solving Agent

Goal-Based Agent

Reason:

A Goal-Based Agent selects actions to achieve a specific goal. In this case, the goal is to transport the customer safely from the pickup location to the destination while considering the selected cab type.

Inputs

- Pickup location
- Destination
- Available cab types
- Fare
- Driver availability
- Estimated arrival time (ETA)

Actions Performed

- Detect pickup location
- Detect destination
- Search available cabs
- Display cab options
- Calculate fare
- Book selected cab
- Assign driver
- Navigate to destination
- Complete payment
- End trip

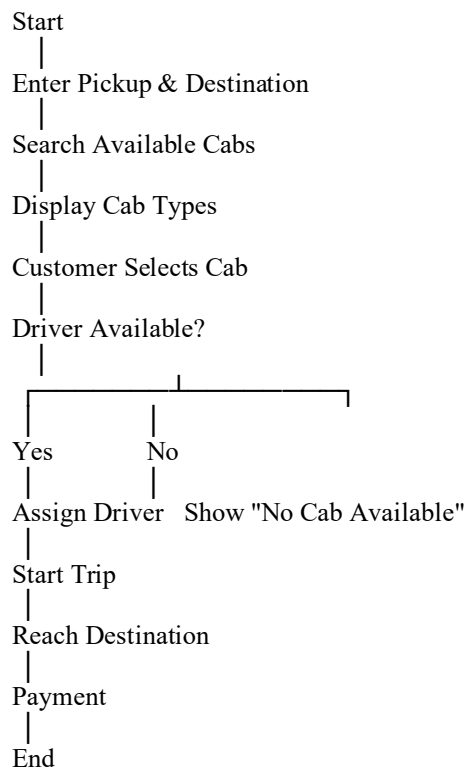
Working of the Agent

1. Customer enters pickup and destination.
2. The system searches nearby available drivers.
3. Different cab options (Mini, Micro, Sedan, Prime, Shared) are displayed.
4. Customer selects the preferred cab.
5. The application calculates the fare and estimated arrival time.
6. The nearest driver is assigned.
7. The driver picks up the customer.
8. The customer reaches the destination.
9. Payment is completed and the trip ends.

Pseudocode

1. BEGIN
2. Input Pickup_Location
3. Input Destination
4. Search Available_Cabs
5. Display:
6. Mini
7. Micro
8. Sedan
9. Prime
10. Share
11. User selects Cab_Type
12. Calculate Fare
13. Estimate Arrival_Time
14. IF Driver Available THEN
15. Assign Driver
16. Confirm Booking
17. Navigate to Destination
18. Complete Trip
19. Make Payment
20. ELSE
21. Display "No Cab Available"
22. ENDIF
23. END

Flowchart



Interpretation of Result

The OLA Cab Booking system functions as a **Goal-Based Intelligent Agent** because it makes decisions based on the customer's objective of reaching the destination efficiently. It searches available options, selects an appropriate cab, assigns a driver, and successfully completes the trip

Question 5: Uniform Cost Search (UCS)

Problem Understanding:

A logistics company needs to transport packages from the **starting warehouse (S)** to the **destination warehouse (G)** with the **minimum travel cost**. The travel cost includes fuel and time. The suitable AI search algorithm is **Uniform Cost Search (UCS)**.

Algorithm Used:

Uniform Cost Search (UCS)

- Finds the least-cost path.
- Expands the node with the lowest cumulative cost first.
- Guarantees the optimal solution when all edge costs are non-negative.

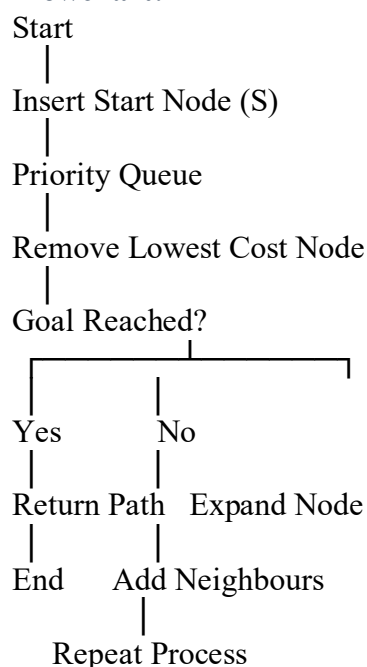
UCS Algorithm:

BEGIN

1. Insert Start node S into the priority queue with cost = 0.
2. Remove the node having the smallest path cost.
3. If the node is Goal (G), stop and return the path.
4. Otherwise, expand the node.
5. Add all neighboring nodes with their cumulative costs.
6. Repeat until Goal is reached.

END

Flowchart:

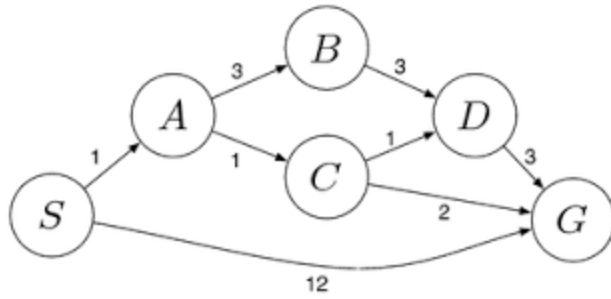


Advantages of UCS:

- Always finds the optimal path.
- Suitable for weighted graphs.
- Widely used in logistics, GPS navigation, and route planning.

Interpretation:

Uniform Cost Search explores paths in increasing order of cumulative cost and guarantees that the first time the destination is reached, it is through the minimum-cost route.



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

